

Ambroise Odonnat

PhD - looking for research scientist positions

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Representation learning and generalization in Transformers

Advisors Ievgen Redko (Huawei), Laetitia Chapel (Institut Agro), Romain Tavenard (Univ. Rennes 2).
Abstract This thesis aims to improve our understanding of representation learning and generalization in transformers through theoretical analysis and large-scale experiments on time series, vision, and language foundation models..
Publications **NeurIPS (2), ICML (3), ICLR (1), AISTATS (1), ICASSP (1), TMLR (2).**
Graduation Januray 2027.

Education

2024–present **PhD in Computer Science, Huawei Noah's Ark Lab & Inria.**
Co-advised by Ievgen Redko, Laetitia Chapel, and Romain Tavenard.
Hosted by Huawei Noah's Ark Lab, Inria, IRISA, and Univ. Rennes 2.
2022–2023 **MSc in Mathematics and Computer Science (MVA), École Normale Supérieure Paris-Saclay.**
Double degree. Advisor: Prof. Gabriel Peyré. **Summa Cum Laude** (GPA: 4.0/4.0).
Main courses: Optimization, Machine learning, Deep learning, Speech & NLP, Computer Vision.
2019–2023 **Engineering Degree, École des Ponts ParisTech, Institut Polytechnique de Paris.**
Advisor: Prof. Gabriel Peyré. **Summa Cum Laude** (GPA: 3.9/4.0).
Main courses: Optimization, Stochastics, Machine Learning, Computer Vision, Computer Science.
2017–2019 **Preparatory Classes (CPGE), Lycée Henri IV, MPSI-MP.**

Research Experience

2024–present **Huawei Noah's Ark Lab, Research Scientist Assistant - CIFRE PhD program.**
○ **Outcome:** developed architectures, benchmarks, and analysis tools to study representation learning and generalization in transformer-based foundation models, including LLMs, VFMs, and TSFMs. Led to **10 publications**, including 5 as a (co)-first author, and **1 product integration**.
○ **Skills:** codebase development, large-scale experiments, precise ablations, theoretical analysis, autonomy, teamwork, communication, open-source contributions, student supervision.
○ **Advisors:** Ievgen Redko, Laetitia Chapel, Romain Tavenard
2023–2024 **Huawei Noah's Ark Lab, Research Scientist.**
○ Developed a transformer model for forecasting ([ICML 2024](#)) and a generalization metric ([TMLR](#)).
○ **Product integration:** [SAMformer](#) model deployed by business units to forecast battery charge cycle.
○ **Advisor:** Ievgen Redko
2023 **Huawei Noah's Ark Lab, Research Intern.**
(6 months) ○ Developed a method for robust self-training under sample selection bias ([AISTATS 2024](#)).
○ **Advisors:** Gabriel Peyré, Ievgen Redko, Vasilii Feofanov
2022 **Polytechnique Montréal, Research Intern.**
(6 months) ○ Developed a deep learning pipeline to detect epileptic spikes on EEG data ([QBIN conference 2022](#)).
○ **Advisor:** Julien Cohen-Adad

Grants and Awards

2025 **ICASSP Oral, Easing Optimization Paths: a Circuit Perspective.**

- 2024 **ICML Oral**, *SAMformer: Unlocking the Potential of Transformers in Time Series Forecasting*.
- 2023 **Merit Scholarship**, *Chair Saint-Gobain & École des Ponts ParisTech*.
- 2022 **Best Flash Talk Award**, *QBIN Conference*.
- 2022 **Research grant**, *Tanenbaum Open Science Institute & McGill University*.

Selected Publications (see my scholar page)

- 2026 **Vision Transformer Finetuning Benefits from Non-Smooth Components**.
ICML - *first author* - vision transformer, finetuning, distribution shifts, PEFT
- 2026 **Optimal Self-Consistency for Efficient Reasoning with Large Language Models**.
ICML - *co-supervision* - LLMs, chain-of-thought, reasoning
- 2026 **Layer by layer, module by module: Choose both for optimal OOD probing of ViT**.
ICLR Workshop - *first author* - vision transformer, distribution shifts, linear probing
- 2026 **Provable Benefits of In-Tool Learning for Large Language Models**.
ICLR Workshop - *first author* - LLMs, tool use, pretraining, SFT, generalization, memorization
- 2025 **Skada-Bench: Benchmarking Unsupervised Domain Adaptation Methods**.
TMLR - *core contributor* - unsupervised domain adaptation
- 2025 **Leveraging Gradients for Unsupervised Accuracy Estimation under Distribution Shift**.
TMLR - *second author* - generalization, distribution shifts
- 2025 **Easing Optimization Paths: a Circuit Perspective**.
ICASSP (Oral) - *first author* - transformer, training dynamics
- 2025 **Clustering Head: A Visual Case Study of the Training Dynamics in Transformers**.
arXiv - *first author* - transformer, training dynamics
- 2024 **Large Language Models as Markov Chains**.
arXiv - *first author* - LLMs, generalization, pretraining, in-context learning - featured on **Forbes**
- 2024 **MaNo: Exploiting Matrix Norm for Unsupervised Accuracy Estimation**.
NeurIPS - *first author* - generalization, distribution shifts - deployed in **Skada**
- 2024 **SAMformer: Unlocking the Potential of Transformers in Time Series Forecasting**.
ICML (Oral) - *first author* - transformers, generalization - deployed in prod. & **PyTorch Forecasting**
- 2024 **Leveraging Ensemble Diversity for Robust Self-Training under Sample Selection Bias**.
AISTATS - *first author* - distribution shifts, self-training, ensemble methods

Invited talks

- Mar. 2026 **Mila**, *Vision transformers finetuning benefits from non-smooth components*, [slides](#).
- Mar. 2026 **Imperial**, *Provable benefits of tool use for large language models*, [slides](#).
- Feb. 2026 **Kyutai**, *Provable benefits of tool use for large language models*, [slides](#).
- Oct. 2025 **Cohere**, *Provable benefits of tool use for large language models*, [slides](#).
- Sep. 2025 **Univ. Lyon 2**, *MaNo: Exploiting matrix norm for unsupervised accuracy estimation*, [slides](#).
- Jun. 2025 **Cohere Labs**, *Large language models as Markov chains*, [slides](#).
- Mar. 2025 **IRISA**, *MaNo: Exploiting matrix norm for unsupervised accuracy estimation*, [slides](#).
- Feb. 2025 **EPFL**, *Large language models as Markov chains*, [slides](#).
- Nov. 2024 **ENS Paris-Saclay**, *Large language models as Markov chains*, [slides](#).
- Sep. 2024 **ENS Ulm**, *SAMformer: transformers for time series forecasting*, [slides](#).
- Sep. 2024 **Telecom**, *Robust self-training under distribution shifts*, [slides](#).
- Sep. 2024 **Inria Soda-Mind**, *Robust self-training under distribution shifts*, [slides](#).
- Jul. 2024 **Criteo**, *SAMformer: transformers for time series forecasting*, [slides](#).

- Jul. 2024 **ENS Paris-Saclay**, *SAMformer: transformers for time series forecasting*, [slides](#).
Jul. 2024 **Inria Malt**, *SAMformer: transformers for time series forecasting*, [slides](#).
Sep. 2024 **Inria Ockham**, *Robust self-training under distribution shifts*, [slides](#).
Feb. 2022 **QBIN Conference**, *Detection of epilepsy on EEG with deep learning*, [slides](#).

Open Source Contributions (see my github page)

- Research Reproducible and open-source code for my articles ([code](#)).
Benchopt Python library for benchmarking ([code](#)).
Skada Python library for domain adaptation ([code](#)).
Skada-bench Domain adaptation benchmark ([code](#)).

Academic services

- 2025 **NeurIPS**, *Workshop Program Chair*.
Co-organized the NeurIPS 2025 workshop “BERT²: Recent advances in time series foundation models”.
2025 **NeurIPS in Paris**, *Program Chair*.
Co-organized [NeurIPS in Paris](#), the local edition of NeurIPS at Sorbonne University (~350 participants).
Reviewing **NeurIPS (2025)**, **ICML (2024, 2025, 2026)**, **ICLR (2025, 2026)**, **AISTATS (2026)**, **TMLR**.

Teaching and supervision

- Intern **Helped create the topic and co-supervised**.
Austin Feng, on efficient reasoning for LLMs ([accepted at ICML 2026](#)).
2024 **Maths examiner**, *HEC-ESSEC competitive entrance exam*.
2024 **Teaching assistant**, *École des Ponts ParisTech*, Machine Learning.
2019–2021 **Maths and physics tutor**, High-school and preparatory classes.

Skills

- Programming Python (7 years), PyTorch (7 years), Shell (3 years)
Tools & OS Git, Linux, Jupyter/Colab, VSCode, LaTeX
Language French (native), English (fluent), German (beginner)

Extra-curricular activities

- 2017 **Concours général Latin-French translation**, *most prestigious French academic competition*.
2015 **Model United Nations in Rome**, *speech towards the General Assembly (~2000 people)*.
Sport Fencing (10 years), Football & Volleyball university team, Tennis

Referees

Charles Arnal

Research Scientist

FAIR at Meta

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Laetitia Chapel

Full Professor

IRISA, Institut Agro Rennes-Angers

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Amaury Hayat

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Ievgen Redko

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Full Professor

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